

Coursework- Econometrics

No AI was used throughout this coursework

1a) FALSE. The seasonal components refer to regular patterns connected to time periods (e.g. short-term fluctuations), whereas cyclical components refer to the rises and falls not of fixed periods (i.e. could be long term fluctuations that do not follow a strict schedule)

b) TRUE. Suppose $Y_t = T_t + S_t + R_t$. You can decompose time series into the trend (T_t), seasonality (S_t) and residual parts (R_t), where the residuals are left after removing seasonality and trend. The trend refers to the long term movement or direction of the data while the seasonality refers to the repeating patterns at regular intervals. Hence the residuals help to analyse any unpredictable variation in the data due to unforeseen events.

c) TRUE. An ECM model is used when variables are cointegrated, meaning they share a long-run equilibrium relationship. The reason for the ECM is that even though the variables are cointegrated and move together over time, in the short run there may be disequilibrium as they may diverge temporarily. Furthermore, according to the Granger Representation Theorem, if cointegration exists between 2 or more variables, an ECM representation exists.

d) TRUE. We can see that this is an MA(3) process.

- An MA(q) process can be represented as $y_t = \mu_t + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} + u_t^-$
- In this case, we have that $\mu_t = 0$, and so $E[y_t] = 0$. $\text{Var}(u_t^-) = 1$
- $\text{Var}(y_t) = \sigma^2 + \sigma^2 \theta_1^2 + \sigma^2 \theta_2^2 + \sigma^2 \theta_3^2$, and so $\text{Var}(y_t) = 1^2 + 0.7^2(1)^2 + 0.2^2(1)^2 + 0.4^2(1)^2 = 1.69$
- We can assume that there is no autocorrelation, i.e. $\text{cov}(y_t, y_{t-k}) = 0$.
Therefore for $k = 0$, $\text{cov}(y_t, y_{t-k}) = \text{cov}(y_t, y_t) = \text{Var}(y_t)$

2a) We notice this is a random walk with drift and so it will not be stationary. For covariance stationarity, we require a constant mean, variance and covariance that does not depend on time t .

$$Y_t = Y_{t-1} + 5 + \varepsilon_t$$

$$\begin{aligned} E[Y_t] &= E[Y_{t-1} + 5 + \varepsilon_t] = E[Y_{t-1}] + E[5] + E[\varepsilon_t] \\ &= E[Y_{t-1}] + 5 = E[Y_0] + 5t, \text{ which depends on time} \end{aligned}$$

$$\text{Var}(Y_t) = \text{Var}[Y_{t-1}] + 1 = \text{Var}(Y_0) + t, \text{ implying variance increases over time.}$$

Therefore Y_t is not covariance stationary

b) We notice an AR(2) process. Hence to assess stationarity we use lag operators.

- $y_t = 0.5 + 0.3L_{t-1} + 0.6L^2y_t + \varepsilon_t$
- $y_t(1 - 0.3L - 0.6L^2) = 0.5 + \varepsilon_t$
- $y_t = \frac{0.5 + \varepsilon_t}{1 - 0.3L - 0.6L^2}$

This is stationary if the roots of the following equation lies outside of the unit circle:

- $1 - 0.3L - 0.6L^2 = 0$
- $L = 1.06497782, -1.56497782$

This satisfies $|L| > 1$ so the equation is outside the unit circle, and hence it is covariance stationarity

c) $E[Y_t] = 0.5E[t] = 0.5t$, which depends on time so it is not covariance stationary.

- $var(y_t) = E((y_t - E(y_t))^2)$
- $E((0.5t + \varepsilon_t + 1.8\varepsilon_{t-1} - 0.5t)^2)$
- $E((\varepsilon_t + 1.8\varepsilon_{t-1})^2)$
- $E(\varepsilon_t^2 + 1.8^2\varepsilon_{t-1}^2 + 1.8\varepsilon_t\varepsilon_{t-1})$
- $E(\varepsilon_t^2) + 1.8^2E(\varepsilon_{t-1}^2) + 1.8E(\varepsilon_t\varepsilon_{t-1})$
- $1 + 1.8^2 = 4.24$, a constant.
- $cov(y_t, y_{t-1}) = E((y_t - E(y_t))(y_{t-1} - E(y_{t-1})))$
- $E((\varepsilon_t + 1.8\varepsilon_{t-1})(\varepsilon_{t-1} + 1.8\varepsilon_{t-2}))$
- $E(\varepsilon_t\varepsilon_{t-1}) + 1.8E(\varepsilon_t\varepsilon_{t-2}) + 1.8E(\varepsilon_{t-1}^2) + 1.8^2E(\varepsilon_{t-1}\varepsilon_{t-2})$
- $= 1.8$, a constant

Since the mean depends on time, the equation is not covariance stationary.

3a) From a theoretical perspective, we would expect these series to be cointegrated. As GDP grows, businesses typically experience higher demand, which encourages increased investment. Additionally, lower interest rates reduce the cost of borrowing, further stimulating investment. Since central banks often adjust interest rates to influence and stabilize GDP growth, these variables are interconnected through a long-run equilibrium relationship.

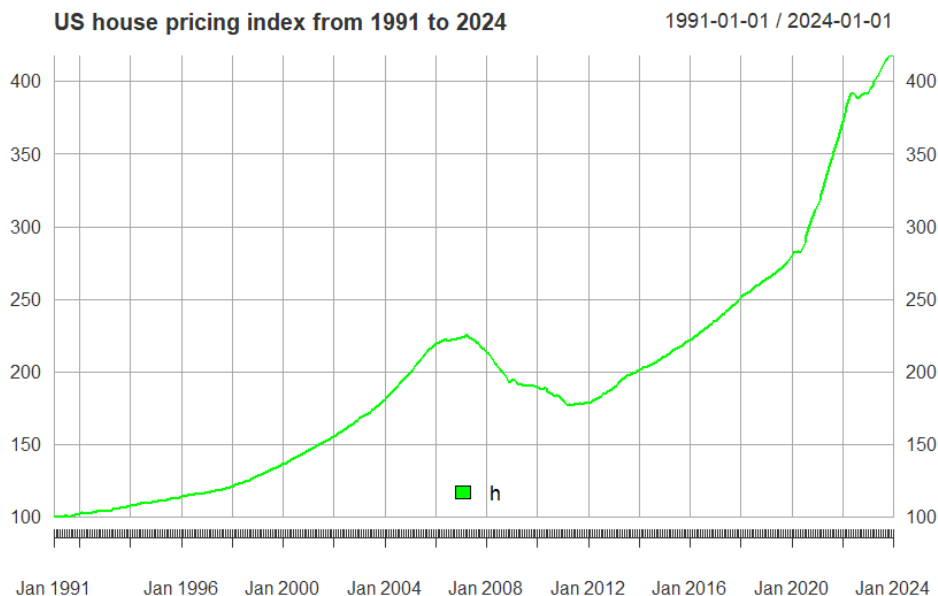
b) We can use an Engle-Granger test in order to test for cointegration.

- Null Hypothesis $H_0 : \gamma = 0$, the residuals are $I(1)$ and there is no cointegration
- Alternate Hypothesis $H_1 : \gamma < 0$, the residuals are $I(0)$ and cointegration exists
- The coefficient of $\hat{u}_{t-1}$ is -0.75 . $-0.75/0.19 = -3.95$
- There are 3 series, we have a 5% significance level and no trend, so the critical value (using table in lecture notes) is $-3.7429 - \frac{8.325}{150} - \frac{13.41}{150^2} = -3.80$

- $-3.95 < -3.80$ and so there is sufficient evidence to reject that null, and we conclude that there is evidence of cointegration.

c) We use the estimated long-run regression, $i_t = 1.2 + 0.55 g_t - 0.30 r_t + u_t$. A 1% increase in GDP increases investment by 0.55%, holding the interest rate constant.

4a)



The plot above shows an overall long-term upward trend, reflecting overall growth in US house prices. From January 1991 to roughly January 2006, there is a steady increase, peaking at around 225, and then there is a sharp decline until 2012, likely due to the 2008 financial crisis. From 2012 onwards, there is a sharp increase in growth, with rapid acceleration after January 2020. The time series does not look stationary, as given the long term upward trajectory, there is a deterministic time trend. The graph does not indicate seasonality as there is no repeating yearly patterns. It is likely that there is a unit root as there seems to be no mean reversion due to the upwards drift and trend. It is likely an ADF test will confirm this.

We conduct an ADF test on the raw data, using type “trend”. We choose to use AIC as we are using slightly longer lag lengths.

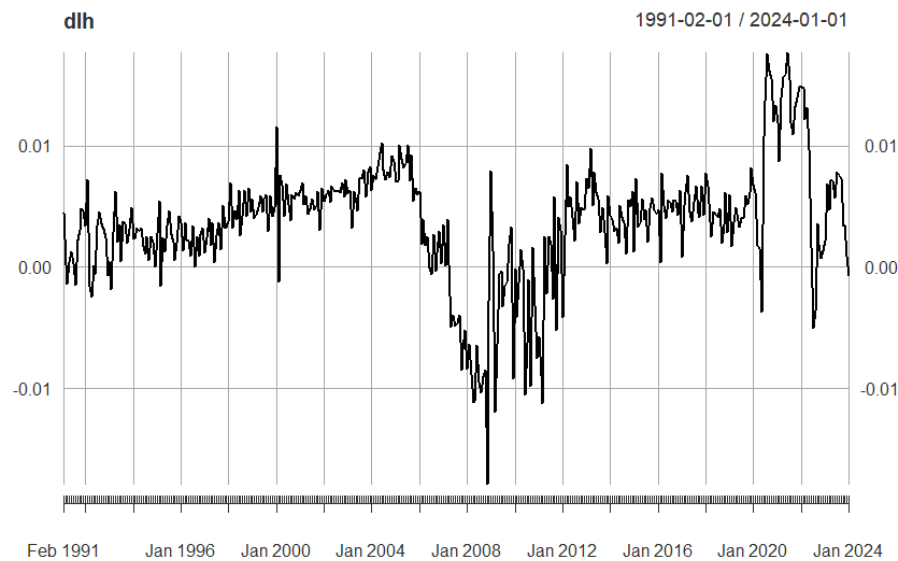
Value of test-statistic is: -1.2123 2.2027 1.6533

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.98	-3.42	-3.13
phi2	6.15	4.71	4.05
phi3	8.34	6.30	5.36

Since we are checking for drift and trend we look at tau3. The test-statistic is -1.2123 which is larger than the 10% critical value of -3.13, and so we fail to reject H_0 and conclude the series has at least one unit root.

To stabilise the variance and make the data suitable for forecasting, we transform it using logarithms and then differencing once.



We can see there are gradual changes in the mean around 0, and so we believe there may be a possibility for drift. We can also see some dynamics in the data, such as the upward and downward movements over multiple periods, so the data does not appear to be white noise but ARMA.

We now conduct an ADF test on the first differenced data, using type “drift”.

Value of test-statistic is: -3.025 4.5756

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.44	-2.87	-2.57
phi1	6.47	4.61	3.79

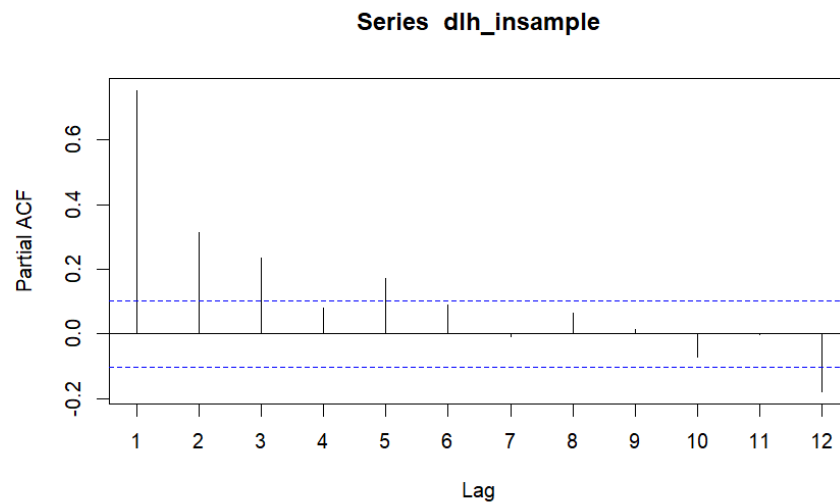
The tau 2 statistic (-3.025) is less than the 5% critical value(-2.87), but more than the 1% value (-3.44), so we reject the null at the 5% level of a unit root. The phi1 comparisons show us that there is drift. The data is now stationary. So we conclude the series is non-stationary in levels, $I(1)$, but after first differencing it is stationary with drift. It is now ready to be used for ARMA modelling.

b)

We define our in-sample period as January 1991 to December 2021 (contains 31years x 12 months = 372 observations) to estimate the model. We then forecast the out-of-sample period as January 2022 onwards (25 observations). This is required so we can compute out of sample errors, and evaluate the model’s predictive accuracy. We would use the best performing model on the entire dataset to project house prices beyond January 2024.

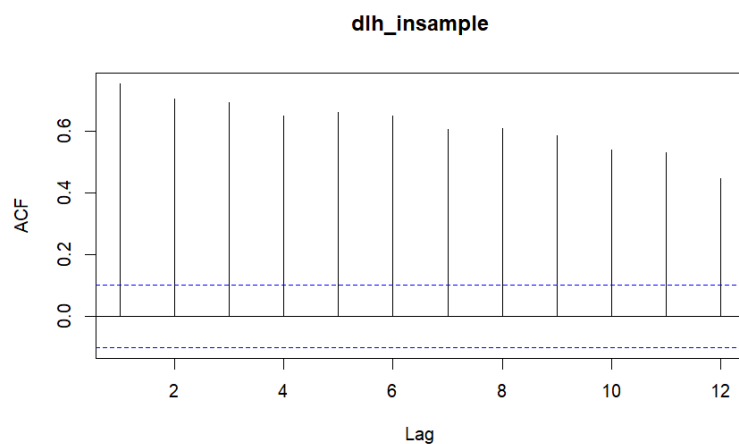
c)

PACF plot:



We can see the auto correlations at lags 1,2,3,5 and 8 are statistically significant.

ACF plot:



The ACF plot shows gradual decay, especially if you increase the number of lags beyond 12, indicating we are likely dealing with an AR(p) process, or that the MA(q) is very low.

d)

Now we look at different ARMA candidates. For models we usually do not have lags past 3. Hence the best candidate models to test are: AR(1), AR(2), AR(3), ARMA(1,1), ARMA(2,1) and ARMA(3,1)

##	sorted AIC	sorted BIC
## 1	ARMA(1,1)	ARMA(1,1)
## 2	ARMA(2,1)	ARMA(2,1)
## 3	ARMA(3,1)	ARMA(3,1)
## 4	ARMA(3,0)	ARMA(3,0)
## 5	ARMA(2,0)	ARMA(2,0)
## 6	ARMA(1,0)	ARMA(1,0)

The table above presents the AIC and BIC values of all the candidate models, and has reordered them from lowest AIC/BIC to highest. We select 2 candidate models to

perform further tests on. We select ARMA(1,1) and ARMA(2,1) as they have the lowest, and second lowest AIC and BIC values.

We now want to assess if the residuals of the 'best' candidate models are homoscedastic, normally distributed and not autocorrelated. A Breusch-Godfrey test helps us to test for autocorrelation of the residual series.

ARMA(1,1):

```
##  
## Breusch-Godfrey test for serial correlation of order up to 5  
##  
## data:  lm(as.vector(residuals11) ~ rep(1, T))  
## LM test = 6.611, df = 5, p-value = 0.2512
```

The null hypothesis is that there is no autocorrelation in the first 5 lags of the residuals. The statistic is LM test = 6.611 and the p-value is 0.2512, so it is not smaller than the 5% critical value. We have very strong evidence that there is no autocorrelation in the residuals.

ARMA(2,1):

```
##  
## Breusch-Godfrey test for serial correlation of order up to 5  
##  
## data:  lm(as.vector(residuals21) ~ rep(1, T))  
## LM test = 4.3473, df = 5, p-value = 0.5006
```

The p-value is 0.5006, and LM test = 4.3473. This is not less than the 5% significance level, so we do not reject the null hypothesis.

We conclude that there is no autocorrelation in the residuals of either candidate.

Next we check for heteroscedasticity of residuals using an ARCH test.

ARMA (1,1):

```
""  
## ARCH LM-test; Null hypothesis: no ARCH effects  
##  
## data:  residuals11  
## Chi-squared = 110.3, df = 12, p-value < 2.2e-16
```

ARMA (2,1):

```
## ARCH LM-test; Null hypothesis: no ARCH effects  
##  
## data:  residuals21  
## Chi-squared = 107.68, df = 12, p-value < 2.2e-16
```

The p-values for both candidates are less than 0.05, and so we reject the null hypothesis that the error terms are homoscedastic, and in turn conclude that there is strong evidence of heteroscedasticity. This is as expected however, given that we are working with small sample sizes, and so we do not reject our models based on this.

We now test for normality of the residuals using a Jarque-Bera test.

```
ARMA(1,1):  ...
            ##  Jarque Bera Test
            ##
            ## data:  residuals11
            ## X-squared = 317.28, df = 2, p-value < 2.2e-16

ARMA(2,1):
            ##  Jarque Bera Test
            ##
            ## data:  residuals21
            ## X-squared = 267.41, df = 2, p-value < 2.2e-16
```

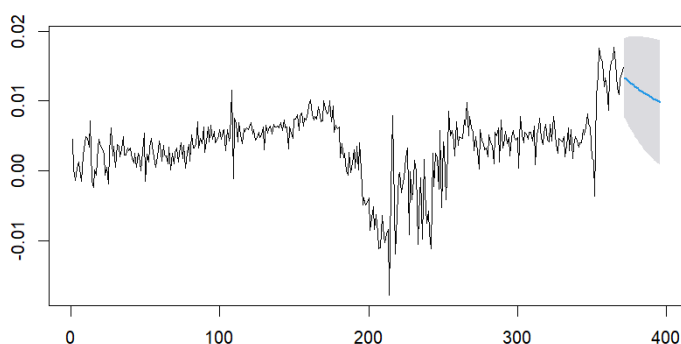
Since the p-values for both candidates are less than 5%, we reject the null hypothesis, meaning the residuals of ARMA(1,1) and ARMA(2,1) are not normally distributed. This is as expected, given our small sample size, and so we do not reject the models based on this test.

Overall we conclude that both ARMA(1,1) and ARMA(2,1) residuals are not autocorrelated, but they are homoscedastic and not normally distributed.

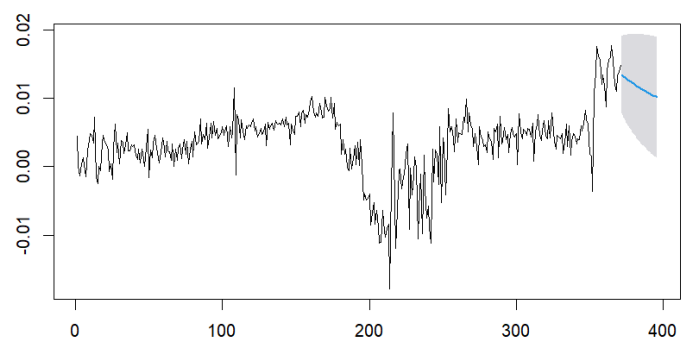
e)

To find the best model, we now forecast the series in the out of sample period, and then based on RMSE and MAE values we will decide which model is best to forecast house prices.

Forecasts from ARIMA(1,0,1) with non-zero mean



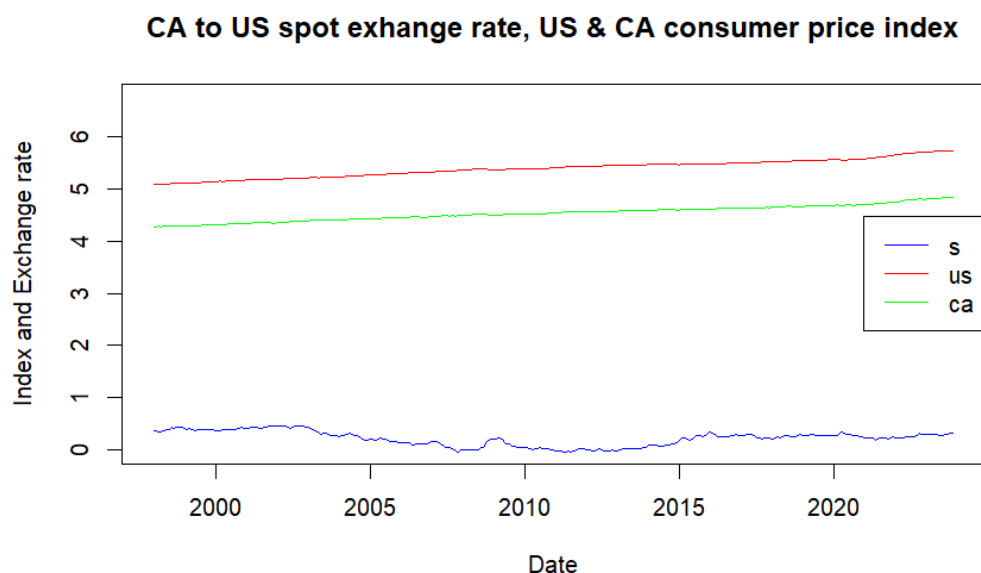
Forecasts from ARIMA(2,0,1) with non-zero mean



	ME	RMSE	MAE	MPE	MAPE	MASE
ACF1						
Training set	2.353419e-05	0.002873162	0.001978039	-Inf	Inf	0.8314322
0.04775675						
Test set	-6.226844e-03	0.007869428	0.006504084	-140.8452	321.977	2.7338719
NA						
	ME	RMSE	MAE	MPE	MAPE	MASE
ACF1						
Training set	2.856527e-05	0.002867616	0.001983084	-Inf	Inf	0.833553
0.006222334						
Test set	-6.428886e-03	0.008038389	0.006688878	-144.9418	330.1937	2.811546
NA						

From the plots, both models look suitable. The model with the lowest RMSE and MAE value is the ARMA(1,1) model. If we were forecasting house prices, using the ARMA(1,1) model, then a RMSE(MAE) of 0.007869428 (0.006504084) would imply a 7.869428 (6.504084) % forecast error, on average.

5a)



From the plot above, we see both US and Canadian Consumer Price Index have steadily increased between 1998 and 2023, reflecting inflation. Based on the Theory of Purchasing Power Parity, which says that in the long run, exchange rates should adjust to equalise the purchasing power of different currencies, i.e. if the US price level rises faster than the Canadian price level, then the CAD should appreciate to maintain parity in real purchasing power, we would expect a long-term relationship. Furthermore, the exchange rate tends not to diverge without bound, suggesting it may be linked to the relative inflation between the countries.

b)

We now apply the Engle-Granger method to test for cointegration between the three series.

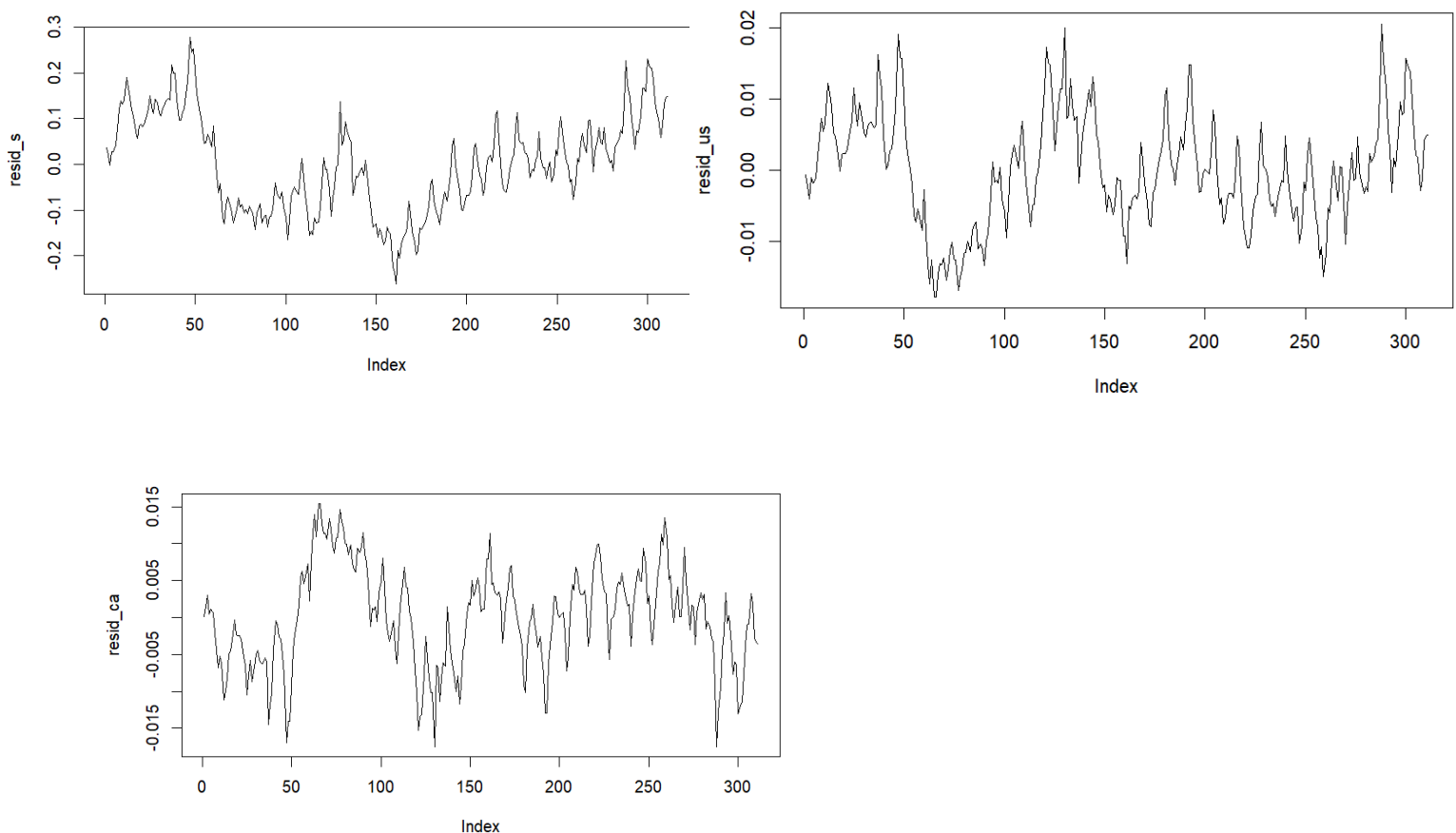
First we do the EG test and ECM, using $s \sim ca + us$, with the Canadian to US Dollar spot exchange rate as the dependent variable. Then we test $us \sim s + ca$, and finally $ca \sim s + us$. We test these different regressions because the ordering of the variables matters.

We see that for all the models, all regressors are highly significant. When we regress exchange rates on the price indexes, we see the R-squared of the regression is 45.27%, but since this regression is spurious, as it involves $I(1)$ variables, all we take from this regression is the residuals. Our critical value from the lecture notes is -3.7429

(asymptotic 5% for the case of 3 variables). The test statistic $-3.1951 > -3.7429$ so it is not significant and we do not reject the null hypothesis that the residuals from the cointegrating equation have a unit root.

This is unlike when we regress US price index on spot exchange rates and Canadian price index, and regress Canadian price index on spot exchange rates and US price index, where r squared is 0.9979, for both.

We next plot the residuals for each regression:



These residual plots above all look stationary. They are not trending and are mean-reverting often. To test for cointegration is to test these residuals for a unit root. We will use the test with drift, so that under the alternative we have a stationary AR with an intercept.

Our critical value from the lecture notes is -3.7429 (asymptotic 5% for the case of 3 variables). The test statistic for regression on spot exchange rate is $-3.1951 > -3.7429$ so it is not significant and we do not reject the null hypothesis that the residuals from the cointegrating equation have a unit root. The test statistics for regression on US price index and Canadian price index is -4.7095 and -4.7262 respectively which is less than

the critical value, and so for these two tests we do reject the null and say that the residuals are stationary and so we have cointegration between the three variables.

Next we estimate an Error Correction Model. According to the Granger Representation Theorem, we need both the result from the EG test, and an error correcting ECM, to verify that we have cointegration.

The form of our ECMs is:

$$\Delta s_t = c + \beta_1 resid_{s,t-1} + \beta_2 \Delta us_t + \beta_3 \Delta ca_t + \varepsilon_t$$

$$\Delta us_t = c + \beta_1 resid_{\{us\},t-1} + \beta_2 \Delta s_t + \beta_3 \Delta ca_t + \varepsilon_{\{us,t\}}$$

$$\Delta ca_t = c + \beta_1 resid_{\{ca\},t-1} + \beta_2 \Delta s_t + \beta_3 \Delta us_t + \varepsilon_{\{ca,t\}}$$

ECM model: $ds \sim lag_resid_s + dus + dca$, where d refers to the data being differenced once.

```
## (Intercept)  0.003476    0.001212    2.868  0.00442 **
## lag_resid_s  0.012141    0.009795    1.239  0.21612
## dus         -1.774163    0.416604   -4.259  2.74e-05 ***
## dca          0.025853    0.337241    0.077  0.93894
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01754 on 306 degrees of freedom
## Multiple R-squared:  0.08939,    Adjusted R-squared:  0.08047
## F-statistic: 10.01 on 3 and 306 DF,  p-value: 2.587e-06
```

We notice that the only significant regressor is dus. The r-squared is low at 0.08047, and so we conclude that this is a spurious regression as all the variables are $I(0)$. The short-run effects of changes in us and ca on s are -1.77 and 0.026 percentage points. If the US CPI goes up by 1 percentage point(pp), the CAD to US spot exchange rate will go down on average by 0-1.77pp, so they move in the opposite directions in the short run. The relationship with the CAD CPI is positive in the short run, if dca goes up by 1pp, s will go up by -0.026pp. Due to Purchasing Power Parity, this is what we expect to see. When US inflation goes up, the US dollar tends to lose value, so you need fewer Canadian dollars to buy one USD, that's why we see a strong negative effect of US inflation on the exchange rate. On the other hand, Canadian inflation doesn't seem to have much of an effect. The impact is small and not statistically significant, which could be because other factors, like interest rates or oil prices, have a bigger influence on the Canadian dollar.

To see if there are long-run effects we need to check whether ds error corrects. The long run relationship is the one we estimated in the cointegrating equation above. We found it to be :

$$s_t = 3.0040 - 8.6951 us_t + 9.7242ca_t$$

The residuals from this regression that are used in the ECM are

$$resid_{s_{t-1}} = s_{t-1} - 3.0040 + 8.6951 us_{t-1} - 9.7242ca_{t-1}$$

If the long run relationship is true in $t - 1$, the residuals will be zero. If there is deviation from the long run relationship, the residuals will be positive or negative. To see if the model is error correcting, we need to see if the coefficient of the resid_s variable, that is the speed of adjustment coefficient. We see that its value is 0.012141con, and p-value is 0.21612, and so it is insignificant, and we do not have error correction. We conclude that these three variables are not cointegrated here.

Now we check if our results hold if we change the ordering.

For $dus \sim lag_resid_us + ds + dca$:

```

...
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0012120  0.0001471   8.239 5.12e-15 ***
## lag_resid_us -0.0538459  0.0179960  -2.992 0.002996 **
## ds          -0.0283562  0.0074930  -3.784 0.000185 ***
## dca           0.4753411  0.0362308  13.120 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002323 on 306 degrees of freedom
## Multiple R-squared:  0.4141, Adjusted R-squared:  0.4083
## F-statistic: 72.08 on 3 and 306 DF, p-value: < 2.2e-16

```

We see that all predictors are significant and the R-squared is 0.4083. The short-run effects of changes in s and ca on us are -0.028 and 0.475 percentage points respectively. If the Canadian CPI goes up by 1 percentage point(pp), the US CPI will go up on average by 0.475, so they move in the same direction in the short run. The relationship with the spot exchange rate is negative in the short run, if ds goes up by 1pp, us will go down by 0.028pp. This is as expected, as Canada and the US are major trading partners with closely linked economies, so inflation shocks in one country often spill over into the other, especially in the short run.

Once again, to check if there are long-run effects, we need to check for error-correction. The long run relationship is the one estimated in the cointegrating equation above. We found it to be:

$$us_t = 0.242983 - 0.046393s_t + 1.137597ca_t$$

The residuals from this regression that are used in the ECM are

$$resid_us_{t-1} = us_{t-1} - 0.242983 + 0.046393 s_{t-1} - 1.137597ca_{t-1}$$

If the long run relationship is true in $t - 1$, the residuals will be zero. If there is deviation from the long run relationship, the residuals will be positive or negative. To see if the model is error correcting, we need to see if the coefficient of the `resid_us` variable, that is the speed of adjustment coefficient. We see that its value is -0.0538459, and p-value is 0.002996, and so it is significant, and we have error correction. If the residuals in $t-1$ are positive, meaning $us(t)$ is too high, or $s(t)$ and $ca(t)$ are too low, then the effect on the short-term change on will be negative and vice versa, showing that error corrects in the short term towards the long-term relationship.

We conclude that the three variables are cointegrated and there is error correction.

For $dca \sim \text{lag_residca} + ds + dus$:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0002271  0.0002049   1.108    0.269

## lag_resid_ca -0.1244934  0.0252448  -4.931 1.34e-06 ***
## ds          -0.0044260  0.0096711  -0.458    0.648
## dus          0.7589589  0.0577609  13.140 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002934 on 306 degrees of freedom
## Multiple R-squared:  0.405, Adjusted R-squared:  0.3992
## F-statistic: 69.44 on 3 and 306 DF, p-value: < 2.2e-16
```

We see that `ds` is not significant while `dus` is. The R-squared value is 0.3992. The short-run effects of changes in s and us on ca are -0.0044 and 0.7590 percentage points respectively. If the US CPI goes up by 1 percentage point(pp), the Canadian CPI will go up on average by 0.7590pp, so they move in the same direction in the short run. The

relationship with the spot exchange rate is negative in the short run, if ds goes up by 1pp, ca will go down by 0.0044pp. This is as expected, due to the strong trade ties and economic integration between the two countries, as well as their shared exposure to global price pressures (e.g., oil, supply chains).

To see if there are long-run effects we need to check if $ca(t)$ error corrects. The long run relationship is the one estimated in the cointegrating equation above,

$$ca_t = -0.202352 + 0.876991s_t + 0.039998us_t$$

The residuals from this regression that are used in the ECM are

$$resid_ca_{t-1} = ca_{t-1} + 0.202352 - 0.876991 s_{t-1} - 0.039998us_{t-1}$$

If the long run relationship is true in $t - 1$, the residuals will be zero. To see if the model is error correcting, we need to see if the coefficient of the $resid_ca$ variable, that is the speed of adjustment coefficient. We see that its value is -0.1244934, and p-value is $1.34e-06$, and so it is significant, and we have error correction. This means that whenever CA CPI at $t-1$ increases, the short-run Δca_t will be negative, pulling CA CPI back down towards the equilibrium path and so a negative residual produces a positive adjustment.

We conclude that the three variables are cointegrated and there is error correction.

c)

The main test is the BG test for autocorrelation, as that will tell us if we need to add more lags to the ECM. We test with 4 lags. The p-value ($3.599e-07$) is less than 5% so the residuals are autocorrelated and we need to add more lags to the model. We create 3 lags for each variable.

After conducting the BG test, we find that all variables are significant, but the most significant variables are dus_3 , ds_2 , ds_3 , dca_1 , dca_2 , dca_3 , and so we remove them from the model.

Our model becomes: $dus \sim lag_resid_us + ds + dca + dus_1 + dus_2 + ds_1$

Now we conduct the BG test again on our new regression and find a p-value of 0.7381, meaning we have strong evidence not to reject the null and conclude that the residuals are not autocorrelated. The differenced lags are the short-term effects and the cointegration is the long-term effects.

